

investment in these 100x50kW power plants will be the same if not less than in a large 5MW plant. The answer is therefore to go for small grid-fed plants owned by small privately-owned utilities. Rs 17 or 18 per unit of buying rate from such plants will attract thousands of small investors like a magnet. In most of the areas in India, solar power can then be a fiscally sound investment that reduces electricity payments immediately, as well as hedges the small solar plant owners against local utility price increases.

Solar Mission's priority, in fact, must be to improve power quality in smaller towns and remote areas. When solar power is fed at 33kV low-impedance point of copper grid, this will not happen! There is also a socio-cultural impact of this. Use of PV solar power is a green initiative on your part, making you eligible for a 'carbon credit.' Carbon credits are a representation of the carbon that would have been emitted if the energy had been generated by a typical fossil fuel power plant instead of a renewable-power plant.

Government's role in encouraging small producers

Change in government policies will help rapid solar power growth in our country via rooftop and other low-power solar plants. Jawaharlal Nehru, a democrat, would have been happy with a faster move to mission targets steered by these new small manufacturers of electricity. I hope the micro, small and medium enterprises (MSME) secretary is reading this. Dr Farooq Abdulah's Ministry of New and Renewable Energy (MNRE) may please quickly clarify when will the power utilities be instructed to buy solar power from 5kW to 100kW solar plants at the same rate as +1MW plants. As more and more rooftop solar power plants feed power into the grid, local power utility companies should be happy to buy power at higher price since this will enable them to own the carbon credits.

MNRE needs to remove all the re-

maining roadblocks to encourage rooftop and backyard solar power plants of 5 to 500kW capacity to feed their solar power into the grid and augment shortage of supply from utility companies. Every other country in the world pays a high rate for such solar power fed into the grid and this makes such investment very well rewarding. USA, Canada, Germany, France, Spain and Portugal not only pay but encourage every citizen to do so. Proven and secure technology to compute the amount of energy in kWh fed into the grid is well developed and widely available even in India. The available metering technology is secure enough for the government not to worry about misuse. Actually, the government need not pay as high as Rs 18. Farooq Saheb can still look generous even with much less!

If a villager to whom the govern-

ment encourage large plants only?

What's in a PV solar power system?

Very soon, the government will be buying power generated from rooftop and backyard solar power plants. Notwithstanding what it decides, let us look at the key considerations in evaluating solar option:

1. Size and physical load capacity of your roof
2. Geographic location of your rooftop or backyard and continuous availability of the sunlight
3. Your own electrical load and current rate
4. State government, local utility, and local community regulations and incentives

Before you go further, it is important to have a basic understanding

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ment has failed to provide access to the electricity generates his own power, say, one unit a day (1 kWh), how can the government not give him Rs 18 for his unit of power if it pays to other city investors in solar generation? He will be well within his rights to demand it than be happy with a few thousand rupees of subsidy given as a favour by an all-powerful government. The government (MNRE) knows well that the investment needed per kilowatt of PV solar power is almost the same for a kW plant installed by the farmer and 10MW plant installed by institutional investors.

Let us remember big plants need big land; about three acres or more per megawatt. That one factor indeed makes such a plant high in cost. It will be a wrong use of such land unless the land is non-arable, or commercially and strategically unsuitable. How unwise it is not to use currently unused millions of rooftops for generating solar power and feeding into the grid

of the components of a solar power system and how it generates electricity.

PV solar power systems are very simple electric power-generating plants comprising the following basic component systems:

1. A set of PV panels that transform sunlight (photons) into DC electricity
2. A racking system that firmly hold the panels to the roof and exposes the panels to the sun at an advantageous angle
3. Inverters that transform the DC electricity into AC electricity
4. Wiring that connects everything
5. Storage battery (In the case of grid-fed power plant, big-size battery is not necessary to store and use that power after sunset.)

Of all the component parts, solar module accounts for the biggest cost, which can be 70 per cent of the total project cost. Cost per watt is currently about Rs 130 to Rs 150 ex-factory. As a result, a variety of solar module types